nuclear force.” The strong force binds together the protons and neutrons in atomic nuclei. Neutrons are electrically neutral—hence the name—but they feel the strong force just like protons. They play a crucial role in the nuclear tug-of-war between electrical repulsion and strong force attraction, aiding the latter without affecting the former. While the strong force isn’t quite strong enough to keep two protons together by itself, adding a neutron to the mix increases the “stickiness” of the strong force without adding any electrical charge, creating a stable atomic nucleus of two protons and one neutron (helium-3).

The nuclear struggle between the sticky strong force and the repellent electrical force ultimately depends on the size of the nucleus. For small nuclei, the strong force wins out easily, and adding more protons and neutrons generally just makes it stronger. But the strong force can only act over very short distances, comparable to the size of a proton itself—anything much larger than a trillionth of a millimeter (a distance known as one fermi, after Enrico) is too much for it. After a certain point, the nucleus gets too big, the electric force starts to win the tug-of-war, and nuclei become weaker as more protons and neutrons are added. Specifically, that point is around nickel (28 protons and 34 neutrons) and iron (26 protons and 30–32 neutrons). Bigger nuclei than that are less stable, and beyond a certain size—namely lead, which has 82 protons and over 100 neutrons—there are no stable nuclei at all.

Uranium is far past that point. With 92 protons, it doesn’t matter how many neutrons you add to uranium—it will eventually decay. But there are two forms of uranium nuclei that will stick around for billions of years before they do: uranium-235 and uranium-238. The numbers refer to the total number of protons and neutrons in the nuclei: U-235 has 143 neutrons and 92 protons, for a total of 235. U-238 has 3 more neutrons, which makes it slightly heavier. But they’re both uranium: the chemical identity of an atomic nucleus is determined solely by the number of protons that it has. Chemistry is all about electromagnetic interactions between atoms. The chemical properties of an atom are entirely determined by the number of electrons it has—and the number of electrons that surround a particular atomic nucleus is determined in turn by the number of protons in that nucleus. Nuclei with the same number of protons but different numbers of neutrons are different *isotopes* of the same element—they differ in weight